

McLennan Derivation for contraction force

~~for one fibre length~~

The force formula which have to derive:

$$F = \frac{\pi d_0^2 P}{4 \sin^2 \theta_0} [(3(1 - \epsilon^2 \cos^2 \theta_0)) - 1]$$

Derive

① $\cos \theta = \frac{L}{b}$... for one fibre length

$$L = b \cos \theta$$

as L is changing so $L_0 \rightarrow$ for initial
 $L_f = L \rightarrow$ for final

$$L = b \cos \theta$$

$$L_0 = b \cos \theta_0$$

$$b = \frac{L}{\cos \theta}$$

By rearranging

$$b = \frac{L_0}{\cos \theta_0} = \frac{L}{\cos \theta}$$

$$b =$$

$$\boxed{\cos \theta = \frac{L}{L_0} \cos \theta_0} \quad \dots \text{①}$$

② Contraction action:

$$\epsilon = \frac{L_0 - L}{L_0}$$

$$L_0 \epsilon = L_0 - L$$

$$L = L_0 - L_0 \epsilon$$

$$L = (1 - \epsilon) L_0$$

$$\frac{L}{L_0} = (1 - \epsilon)$$

$$\boxed{\cos \theta = (1 - \epsilon) \cos \theta_0} \quad \dots \text{②}$$

by trigonometric relation
 $\sin^2 \theta + \cos^2 \theta = 1$

as

$$\cos \theta = (1 - \epsilon) \cos \theta_0$$

$$\cos^2 \theta = (1 - \epsilon)^2 \cos^2 \theta_0$$

$$\sin^2 \theta + (1 - \epsilon)^2 \cos^2 \theta_0 = 1$$

$$\sin^2 \theta = 1 - (1 - \epsilon)^2 \cos^2 \theta_0$$

$$\sin \theta = \sqrt{1 - (1 - \epsilon)^2 \cos^2 \theta_0}$$

③ As instantaneous length

$$L = L_0 (1 - \epsilon)$$

since

$$b = \frac{L}{\cos \theta} = \frac{L_0}{\cos \theta_0}$$

$$\therefore b = L = \frac{L_0 \cos \theta}{\cos \theta_0}$$

similarly using $b = \frac{\pi d}{\sin \theta}$

the instantaneous diameter d is

$$d = \frac{b \sin \theta}{\pi} \quad d = \frac{L_0 \sin \theta}{\pi \sin \theta_0}$$

$$d = \frac{L_0 \sin \theta}{\sin \theta_0}$$

$$\frac{\pi d_0}{\sin \theta_0}$$

④ Put eqⁿ ③ in eqⁿ ④

$$d = \frac{L_0}{\sin \theta_0} \sqrt{1 - (1 - \epsilon)^2 \cos^2 \theta_0}$$

$$b = \frac{\pi d}{\sin \theta}$$

$$d = \frac{b \sin \theta}{\pi}$$

$$d = \frac{L_0 \sin \theta}{\sin \theta_0}$$

$$d = \frac{L_0 \sin \theta}{\sin \theta_0}$$

⑤ Volume:

$$V = \frac{\pi d^2}{4} L$$

$$\therefore L = (1 - \epsilon) L_0$$

$$L_0 = \frac{\pi d_0}{\tan \theta_0}$$

$$L = (1 - \epsilon) \frac{\pi d_0}{\tan \theta_0}$$

$$V = \frac{\pi d^2}{4} (1 - \epsilon) \frac{\pi d_0}{\tan \theta_0} \quad \text{--- ⑥}$$

Put eqⁿ ③ in ⑥

$$V = \frac{\pi}{4} \left[\frac{d_0^2}{\sin^2 \theta_0} \times (1 - (1 - \epsilon)^2 \cos^2 \theta_0) \right] \frac{\pi d_0}{\tan \theta_0} (1 - \epsilon)$$

$$V = \frac{\pi^2 d_0^3 (1 - \epsilon)}{4 \tan \theta_0} \left[\frac{1}{\sin^2 \theta_0} [1 - (1 - \epsilon)^2 \cos^2 \theta_0] \right]$$

$$V = \frac{\pi^2 d_0^3 (1 - \epsilon)}{4 \tan \theta_0} \left[\frac{1}{\sin^2 \theta_0} - \frac{(1 - \epsilon)^2 \cos^2 \theta_0}{\sin^2 \theta_0} \right]$$

$$V = \frac{\pi^2 d_0^3 (1 - \epsilon)}{4 \tan \theta_0} \left[\frac{1}{\sin^2 \theta_0} - (1 - \epsilon)^2 \cot^2 \theta_0 \right]$$

$$V = \frac{\pi^2 d_0^3 (1 - \epsilon)}{4 \tan \theta_0} \left[(1 + \cot^2 \theta_0) - (1 - \epsilon)^2 \cot^2 \theta_0 \right]$$

$$V = \frac{\pi^2 d_0^3 (1 - \epsilon)}{4 \tan \theta_0} \left[\cot^2 \theta_0 \left[\frac{1}{\cot^2 \theta_0} + 1 \right] - (1 - \epsilon)^2 \cot^2 \theta_0 \right]$$

$$V = \frac{\pi^2 d_0^3 (1 - \epsilon)}{4 \tan^3 \theta_0} \left[\sec^2 \theta_0 - (1 - \epsilon)^2 \right] \quad \text{--- ⑦}$$

⑥ The circumferential force:

$$T = (F + AP) \tan \theta$$

$$T = \left(F + \frac{\pi d^2 P}{4} \right) \tan \theta$$

∴ Radial force

$$P \frac{d}{2} d\phi$$

∴ Sum of components of the force applied on circumference

$$2T = 2 \int_0^{\pi} P \frac{d}{2} \sin\phi d\phi$$

$$2T = Pd \left[-\cos\phi \right]_0^{\pi}$$

$$2T = Pd \left[-\cos\frac{\pi}{2} - (-\cos(0)) \right]$$

$$2T = Pd [-0 + 1]$$

$$2T = Pd$$

$$T = \frac{Pd}{2} \quad \dots (8)$$

As, $T = \left(F + \frac{\pi d^2 P}{4} \right) \tan\theta$

$$\frac{Pd}{2} = \left(F + \frac{\pi d^2 P}{4} \right) \tan\theta$$

$$\frac{Pd}{2 \tan\theta} = F + \frac{\pi d^2 P}{4}$$

$$\therefore \tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\sqrt{1 - (1-\epsilon)^2 \cos^2\theta_0}}{(1-\epsilon) \cos\theta_0}$$

$$d = \frac{d_0}{\sin\theta_0} \sqrt{1 - (1-\epsilon)^2 \cos^2\theta_0}$$

$$\frac{Pd}{2 \tan\theta} = \frac{P}{2} \times \left[\frac{\frac{d_0}{\sin\theta_0} \sqrt{1 - (1-\epsilon)^2 \cos^2\theta_0}}{\sqrt{1 - (1-\epsilon)^2 \cos^2\theta_0}} \right]$$

$$\frac{Pd}{2 \tan\theta} = \frac{Pd_0 (1-\epsilon) \cos\theta_0}{2 \sin\theta_0}$$

$$F = \frac{Pd}{2 \tan\theta} - \frac{\pi d^2 P}{4}$$

$$\frac{\pi d^2 P}{4} = \frac{P\pi}{4} \left[\frac{d_0^2}{\sin^2\theta_0} (1 - (1-\epsilon)^2 \cos^2\theta_0) \right]$$

$$\frac{\pi d^2 P}{4} = \frac{P\pi}{4} \left[\frac{d_0^2}{\sin^2\theta_0} - \frac{d_0^2}{\sin^2\theta_0} (1-\epsilon)^2 \cos^2\theta_0 \right]$$

$$\frac{\pi d^2 P}{4} = \frac{\pi d_0^2 P}{4 \sin^2\theta_0} \left[1 - (1-\epsilon)^2 \cos^2\theta_0 \right]$$

$$\frac{Pd}{2 \tan\theta} = \frac{P}{2 \sin\theta_0} (1-\epsilon) \cos\theta_0$$

$$\therefore \cos\theta = (1-\epsilon) \cos\theta_0$$

$$\frac{\pi d^2 P}{4} = \frac{\pi d_0^2 P}{4 \sin^2\theta_0} [1 - \cos^2\theta]$$

$$\frac{Pd}{2 \tan\theta} = \frac{Pd_0 \cos\theta}{2 \sin\theta_0}$$

for extracting out the term $\frac{\pi d_0^2 P}{4 \sin^2\theta_0}$

$$\frac{\pi d_0^2 P}{4 \sin^2\theta_0} (1-\epsilon) \cos\theta_0 \sin\theta_0$$

$$\frac{Pd}{2 \tan\theta} = \frac{\pi d_0^2 P}{4 \sin^2\theta_0} \left[\frac{2 \sin\theta_0}{\pi d_0} (1-\epsilon) \cos\theta_0 \right]$$

$$= \frac{\pi d_0^2 P}{4 \sin^2\theta_0} \left[\frac{2 \sin\theta_0 \cos\theta}{\pi d_0} \right]$$

$$F = \frac{\pi d_0^2 P}{4 \sin^2\theta_0} \left[\frac{2 \sin\theta_0 \cos\theta}{\pi d_0} - (1 - \cos^2\theta) \right]$$

$$F = \frac{\pi d_0^2 P}{4 \sin^2\theta_0} \left[\frac{2 \sin\theta_0 \cos\theta}{\pi d_0} - 1 + \cos^2\theta \right]$$

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$$F = \frac{\pi d_0^2 P}{4 \sin^2 \theta_0} \left[\frac{2 \sin \theta_0 \cos \theta + \cos^2 \theta - 1}{\pi d_0} \right]$$

$$F = \frac{\pi d_0^2 P}{4 \sin^2 \theta_0} \left[\cos \theta \left[\frac{2 \sin \theta_0}{\pi d_0} + \cos \theta \right] - 1 \right]$$

$$\therefore \cos \theta = (1-E) \cos \theta_0$$

$$F = \frac{\pi d_0^2 P}{4 \sin^2 \theta_0} \left[(1-E) \cos \theta_0 \left[\frac{2 \sin \theta_0}{\pi d_0} + (1-E) \cos \theta_0 \right] - 1 \right]$$

$$b = \frac{\pi d_0}{\sin \theta_0}$$

$$\frac{2}{b} +$$

$$F = \frac{\pi d_0^2 P}{4 \sin^2 \theta_0} \left[(1-E)^2 \cos^2 \theta_0 \left[\frac{2 \cos \theta_0 (1-E)}{b} + 1 \right] - 1 \right]$$

$$\frac{\cos \theta_0 (1-E)}{b} = 1$$

Paper:

Motion Analysis of McKibben Type Pneumatic Rubber Artificial Muscle with Finite Element Method

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This study aims to use three-dimensional (3D) Finite Element Modeling (FEM) to establish a quantitative design optimization method for McKibben-type pneumatic rubber artificial muscle. First, a simple 3D model that does not account for the friction between the tube and the fiber braid strands and or that between the strands themselves is developed. The model is validated through experimentation, and the usefulness of the model is examined. With this model, the effects of various parameters, e.g., the braid angle, on the operation of the artificial muscle is investigated. It is found that the characteristics of the artificial muscle can be predicted. Thus, the proposed analysis may be a useful design method for braided artificial muscles.

Keywords: McKibben type pneumatic rubber artificial muscle, artificial muscle, finite element method, solid model, soft actuator

1. Introduction

1.1. McKibben Type Pneumatic Rubber Artificial Muscle

McKibben type pneumatic rubber artificial muscle, made of rubber and pneumatically activated, is a well-known soft actuator.

McKibben type pneumatic rubber artificial muscle has a rubber tube covered with interlaced fiber strands. Radial expansion forces to be generated by increasing the internal pressure of the rubber tube are converted into axial contraction forces by the interlaced fiber strands. The principle of such McKibben type pneumatic rubber artificial muscles was developed for the first time in the U.S.A. in 1958 for possible applications to artificial legs. However, it was never put into practical use because of its short lifetime and large nonlinearity [1]. Later on, in the middle 1980s, a domestic manufacturer marketed such artificial muscles under the trade name of "Rubbertuator." It

drew so many people's attention at first that they analyzed its characteristics and studied its control methodology for possible applications to painting robots, window cleaning robots, nursing care robots, etc.

It, however, did not meet the requirements for the high-speed, high-accuracy positioning and speed-controlling performance required for ordinary industrial robots. Consequently, the domestic manufacturer does not produce or sell it any more. At present, similar artificial rubber muscles are available from British and German manufacturers in limited sizes and specifications and for considerably high prices.

On the other hand, we have demonstrated that McKibben type rubber artificial muscles can easily be produced at affordable prices using commercially available rubber tubes and interlaced fiber strands. The prototype McKibben type rubber artificial muscle we have developed uses a rubber tube, 11.6 mm in outer diameter, 8.0 mm in inside diameter, and 793 mm in natural length. Its contraction coefficient reaches a saturation point at around 25%. It depends so much on the braid angle of the interlaced fiber strands that it becomes larger when the axial braid angle of the strands in natural length is smaller. When the pneumatic rubber artificial muscle is pressurized up to 600 kPa, a contraction force of approximately 340 N is generated. Such generated forces will increase with longer natural lengths of the interlaced fibers. Those characteristics roughly match the performance of already commercialized artificial muscles. Moreover, the developed McKibben type pneumatic rubber artificial muscle can be handmade in various sizes so easily that it can be applied in a variety of areas.

It also has the following characteristics:

- It has passive compliance.
- Its light weight and high power guarantee a very large power-mass ratio.
- It has so few sliding areas that it generates less friction than any ordinary pneumatic cylinder.
- It requires no lubrication and has no adverse effects on its surroundings.

1. This paper partially contains the translation of the article published by the Japan Society of Mechanical Engineers (JSME).



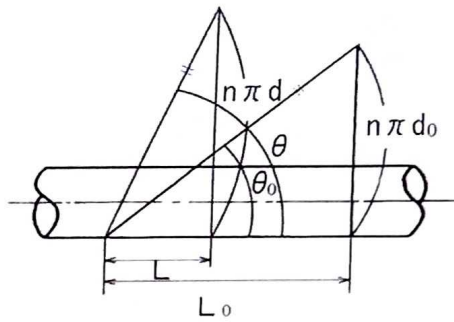


Fig. 3. Geometry of braids.

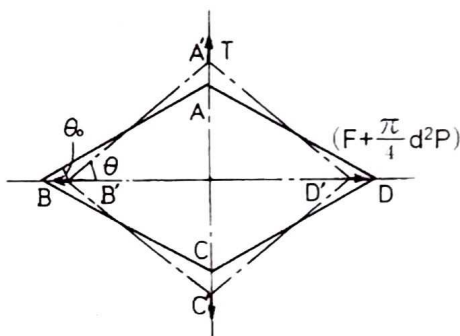


Fig. 4. Braid angle after expansion.

in Fig. 3 that its geometrical constraints vary its overall length instead.

As the fiber length per turn remains unchanged before and after it contracts, its diameter d can be expressed as follows:

$$d = d_0 \frac{\sin \theta}{\sin \theta_0} \quad (1)$$

Figure 4 schematically shows how the braid angle of the interlaced fiber strands changes when the McKibben type pneumatic rubber artificial muscle expands.

Contraction of the McKibben type pneumatic rubber artificial muscle deforms a parallelogram ABCD in the interlaced fiber into a parallelogram A'B'C'D'. This makes the following formula:

$$\frac{L_0}{\cos \theta_0} = \frac{L}{\cos \theta} \quad (2)$$

Then,

$$\cos \theta = \frac{L}{L_0} = (1 - \varepsilon) \cos \theta_0 \quad (3)$$

$$\sin \theta = \{1 - (1 - \varepsilon)^2 \cos^2 \theta_0\}^{\frac{1}{2}} \quad (4)$$

The following equation is obtained by substituting

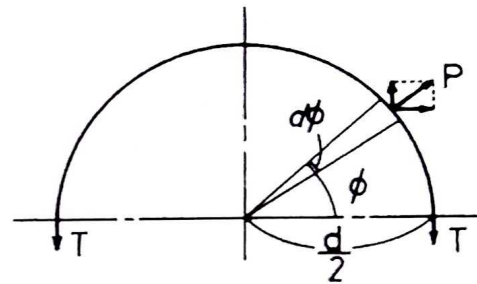


Fig. 5. Circumferential expansion ratio.

Eq. (4) into Eq. (1).

$$d = \frac{d_0}{\sin \theta_0} \{1 - (1 - \varepsilon)^2 \cos^2 \theta_0\}^{\frac{1}{2}} \quad (5)$$

As volume V for the length L of one fiber turn is expressed as follows,

$$V = \frac{\pi d^2 L}{4} \quad (6)$$

$$L = (1 - \varepsilon) L_0 = \frac{\pi d_0}{\tan \theta_0} (1 - \varepsilon) \quad (7)$$

The following equation is obtained by substituting Eqs. (5) and (7) into Eq. (6).

$$V = \frac{\pi^2}{4} \left(\frac{d_0}{\tan \theta_0} \right)^3 (1 - \varepsilon) \left\{ \frac{1}{\cos^2 \theta_0} - (1 - \varepsilon)^2 \right\} \quad (8)$$

We now discuss the balance of forces in the upper half of the McKibben type pneumatic rubber artificial muscle as illustrated in Fig. 5.

The circumferential expansion force T is expressed as follows:

$$\left(F + \frac{\pi}{4} d^2 P \right) \tan \theta = T \quad (9)$$

In the unit length of the artificial muscle, the radial force applied to its minute parts is expressed as follows:

$$P \frac{d}{2} d\phi \quad (10)$$

The sum of the vertical components of the forces applied to the curved part of the semicircle is expressed as follows:

$$2T = 2 \int_0^{\frac{\pi}{2}} P \frac{d}{2} \sin \phi d\phi = Pd \quad (11)$$

Thus, we can establish from Eqs. (5), (9), and (11) the relationship between contraction forces F of the artificial muscle, its internal pressure P , and its contraction coefficient ε , as expressed by Eq. (12).

$$F = \frac{\pi d_0^2 P}{4 \sin^2 \theta_0} \{3(1 - \varepsilon)^2 \cos^2 \theta_0 - 1\} \quad (12)$$

As for the principle of operation, we have reviewed the conventionally advocated theoretical formula to reestablish contraction force

$$P = \frac{F \times 4 \sin^2 \theta_0}{\pi \times d_0^2 \times [3(1 - \varepsilon)^2 \cos^2 \theta_0 - 1]}$$

$$d = \frac{d_0 \sin \theta}{\sin \theta_0}$$

← final braid angle
← initial braid angle
← initial Diameter
← final Diameter

Internal pressure

Contraction ratio

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$$L_0 = b \cos \theta_0$$

we obtain

$$(1 - \varepsilon) \cos \theta_0 = \cos \theta$$

Therefore

$$\frac{2(1 - \varepsilon) \cos \theta_0}{b} = 2 \cos^2 \theta$$

Finally,

$$\cos^2 \theta = (1 - \varepsilon)$$

Thus

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$$F = \frac{\pi d_0^2 P}{4 \sin^2 \theta_0} \left[3(1 - \varepsilon)^2 \cos^2 \theta_0 - 1 \right] .$$

The standard theoretical contraction force (F) of a McKibben pneumatic artificial muscle is calculated using the **Chou-Hannaford / Schulte**

formula: $F = \frac{\pi D_0^2 P}{4} [3\cos^2(\theta) - 1]$ or expressed

via the contraction ratio.  IAEME Publication +2